Adding Ridge Regularizer to the model λ=10

Train

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	11
X_2	1	3	5	6	16	7	5	3
y	2	4	6	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 0.13 \\ 1.87 \\ 0.09 \end{pmatrix}$$

$$\|W\|^2 = 3.5$$

Test

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	57.4	76.5	95.6

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	30
X_2	1	3	5	6	16	7	5	3
y	2	4	30	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 0.54 \\ 0.73 \\ 0.81 \end{pmatrix}$$

$$\|W\|^2 = 1.48$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	34.6	46.4	57.4

	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
X_1	1	2	3	4	8	9	10	60
X_2	1	3	5	6	16	7	5	3
y	2	4	60	8	16	18	20	22

$$\underline{W} = \begin{pmatrix} 0.82 \\ 0.33 \\ 1.1 \end{pmatrix}$$

$$\|W\|^2 = 1.92$$

	$Test_1$	$Test_2$	$Test_3$
X_1	30	40	50
X_2	15	20	25
y	26.8	35.4	44.1